

Pick the wave format

VCD is familiar and portable for quick debug

Scenario-first choice

- For quick debug on a tiny block, VCD is often fine
- For overnight farms with millions of cycles, prefer FST to keep disks sane
- When you share a snippet with a colleague on another tool
- Regardless of format

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Wave dump literacy

Document-aid literacy: **VCD** vs **FST** roles for dumping waves (not GTKWave itself).
Starter: long farm run → FST — MATCHED.

Starter example: scenario long_farm → format FST with dump on — MATCHED.

Load starter example

Challenges 0 / 22 cleared

Quiz: **VCD**: VCD is best described as...

☐ a portable text value-change dump — larger, easy to share/grep

☐ a place-and-route database

☐ GTKWave itself

☐ only a Verilator C++ header

Show hint

Check

Next

Idle

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Core ideas

VCD

Text dump — portable, larger.

FST

Compact binary — long / wide runs.

dump on

Must enable before a file appears.

MATCHED

Format fits the dump scenario.

Lab

SCENARIO PACK

Lab starter

learn_verilator — wave dump

Real Verilator practice

- In Track A, produce one small VCD and, if your flow supports it, one small FST from the
- Compare file sizes and open both in your viewer
- Note which Makefile variables or flags differ
- Keep runs short, this is format literacy, not volume testing

Pitfalls to watch

- Do not pick FST then forget trace-fst at compile
- Do not share a zero-byte file because dump never ran
- Do not use farm-scale dumps for a five-signal typo debug
- And do not assume every viewer supports every format, check before you send

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, pass scenario chooser challenges with dump truly enabled
- In Track A, open one real trace you created
- When you are ready, take the short quiz, then continue to Verilator public visibility